

An untrained escape circuit from the *Drosophila* connectome plays pinball

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Abstract

Connectome-constrained models ask whether the measured wiring of a nervous system carries computational meaning, but most are evaluated on abstract benchmarks. We embed subnetworks of the *Drosophila* male CNS connectome (MaleCNS v1.0) [1] in a closed loop with a decompiled 3D Pinball Space Cadet engine [2], and use degree-preserving shuffles of the same graph as controls to ask whether the real wiring matters. We test two routes. In the first, a trained linear readout sits on a fixed 444-neuron sensory-to-motor subnetwork. It plays about as well as a hand-written reflex, but brains trained on three shuffled wirings do equally well, so training absorbs the wiring. In the second, nothing is trained. We take the looming-escape pathway, in which LC4 and LPLC2 visual neurons project strictly ipsilaterally onto the DNp01 giant fibers, and use it as a body that maps ball approach to flipper presses. Under a matched press budget of at most 25 presses per life, the real giant-fiber body scores 11.221 log1p points per ball life, against 10.770–10.985 for six degree-preserving shuffles (pooled difference -0.338 ± 0.079 , $t = -4.25$). It gets there by pressing more efficiently, not more often, and it stays 0.26 below a one-threshold reflex. The results suggest that wiring matters most when nothing is trained, and that learning can substitute for it. Three caveats apply: the framing is post hoc, the sensory transduction is hand-designed, and there is no side-preserving shuffle control.

1 Introduction

Complete insect connectomes now make it possible to build models whose connectivity is measured rather than fitted. Whether that wiring does any work in an embodied task is rarely tested. The alternative is that any graph of the same size and degree distribution would perform equally well once the free parameters are fit. One direct test runs the same model on the real graph and on randomised graphs that preserve basic statistics, under an identical protocol.

We ask this in a deliberately simple closed-loop task. A fruit-fly connectome subnetwork controls the two flippers of a pinball table, and the score is the points earned in a single ball life. We compare two routes. The first is conventional: take a fixed 444-neuron sensory-to-motor subnetwork and train a small readout on top of it. The second removes training. The fly already has a circuit whose job is to turn an approaching object into a fast motor command, the looming-escape pathway from LC4 and LPLC2 visual neurons to the DNp01 giant fibers [3, 4, 5, 6]. If each flipper is treated as an “eye” and the ball as an approaching object, the pathway can serve directly as a body that decides when to press.

We contribute a closed-loop benchmark in which connectome subnetworks control a real game engine, with one-ball-life episodes and paired seeds for exact comparisons. The first route gives a negative result: a trained readout on a 444-neuron subnetwork shows no advantage of real wiring over three degree-preserving shuffles. The second gives a positive one: an untrained giant-fiber body built from 373 MaleCNS cells beats six degree-preserving shuffles under a matched press budget, with the advantage located in press efficiency. Two diagnostics accompany these results. The real circuit is markedly more recurrent than its shuffles and needs sub-critical scaling to

avoid latching, and it responds to one side only, while shuffles respond on both. We also report the limitations, including a post hoc hypothesis, a hand-designed sensory transduction, and a measured blind spot between the flippers.

2 Methods

2.1 Task and environment

The environment is a decompilation of 3D Pinball for Windows – Space Cadet [2], run headless with a deterministic engine per seed, so any two policies can be compared on identical games (paired evaluation). The agent observes ball position and velocity and the flipper states, and outputs two binary actions, one per flipper. Each decision is held for four engine ticks (the engine runs at 120 ticks per second in the browser, so one decision every 4 ticks).

An episode is one ball life. It ends at the first detected drain or at a step cap (3000 steps for the evaluations here). The score is $\log_{1p}$ of the points earned during that life, reported as mean $\pm$ standard error (SE) over a set of seeds. The game does not penalise presses, which matters for the controls below. Action 0 (left flipper) is at $+x$, because the table x axis is mirrored relative to the screen.

2.2 Connectome and circuit dynamics

Connectivity comes from MaleCNS v1.0 [1] (166,691 annotated neurons, 11,691 cell types). Each edge carries a synapse count, and its sign follows the presynaptic cell’s consensus neurotransmitter (acetylcholine +1, GABA and glutamate −1, others 0). The weight of an edge is its signed synapse count divided by the total unsigned input to the postsynaptic cell, so each cell’s total unsigned input is normalised to 1.

Dynamics are a discrete-time leaky-tanh rate network. Each decision step runs three inner iterations of

$$a \leftarrow (1 - 0.7) a + 0.7 \tanh(\text{drive} + g W a),$$

where the leak is 0.7, the default gain g is 1.4, W is the fixed signed normalised weight matrix, and the drive is non-zero only on input cells. There is no spiking, conductance, synaptic delay or cell-specific time constant.

2.3 Trained 444-neuron brain

Circuit. Input cells are `vnc_sensory` cells ranked by direct weight onto `vnc_motor` cells and split into rank tiers, one tier per observation channel. Outputs are `vnc_motor` cells, and interneurons are two-hop bridge cells. The circuit has 444 nodes (180 input cells from 15 channels of 12 cells, 200 interneurons, 64 output cells) and 7,267 edges. It is never trained.

Encoding. Each ball variable is spread over its 12 sensory cells instead of being broadcast as a scalar (“popcode”). Ball y gets Gaussian bumps, ball x tiles the table width, and each velocity component becomes two sets of 6 direction-selective sigmoid units with graded speed thresholds. This resembles range fractionation by proprioceptors, though the assignment of cell subtypes to channels remains an engineering choice. The agent sees the flipper-state channels one decision step late (“delay1”). With raw flipper feedback, the circuit became a free-running period-3 press/release oscillator that ignored the ball.

Readout. The 667 trained parameters are a per-cell standardisation of the 64 readout cells (128), a bias-free $64 \rightarrow 8$ linear bottleneck (512), and an $8 \rightarrow 3$ head with bias (27).

Training. CMA-ES [7] on whole-life fitness failed to learn ball timing. The best CMA champion was a metronome: left and right presses were identical on 99% of steps, 98% of press gaps were exactly 3 steps, and it made 862 presses per life while staying essentially blind to the

ball. The objective does not penalise presses, so an open-loop rhythm scores near the optimum. We therefore trained by DAgger [8] imitation of a hand-written lead reflex that presses when $y + v_y \cdot 0.0347 > 11.5$ (the “lead1” reflex). Pressing one decision step earlier tripled multiplier-bank hits per approach (0.051 vs 0.016) without more drains. The same pipeline was run on each shuffled wiring.

2.4 Giant-fiber body

Pathway analysis. In the full MaleCNS, LC4 and LPLC2 connect to DNp01 strictly ipsilaterally, with zero synapses to the contralateral giant fiber. Direct synapses and share of the giant fiber’s total input are:

Eye cells $\rightarrow$ giant fiber	cells	synapses	share of GF input
LC4 L $\rightarrow$ GF L	71	3,782	15.2%
LC4 R $\rightarrow$ GF R	55	2,580	13.9%
LPLC2 L $\rightarrow$ GF L	94	2,642	10.6%
LPLC2 R $\rightarrow$ GF R	91	2,220	11.9%
any contralateral pair	–	0	0%

Table 1: Direct synapses from LC4 and LPLC2 onto the DNp01 giant fibers.

Together LC4 and LPLC2 provide 25.8% of each giant fiber’s input. The real circuit is therefore a lateralised looming detector, with the left giant fiber (GF_L) driving the left flipper and GF_R the right.

Circuit. The body contains all LC4 (126) and LPLC2 (185) cells (311 input cells), the two DNp01 giant fibers, 40 top two-hop intermediates (LC4/LPLC2 $\rightarrow$ X $\rightarrow$ GF) and 20 top other direct GF inputs. That makes 373 cells, 15,053 edges and 134,680 synaptic contacts (minimum 3 synapses per edge). Input cells are assigned to a left or right channel by soma side. The body is not trained.

Sub-critical scaling. With the default gain, the real circuit has spectral radius $\rho = 0.996$, so $\text{gain} \times \rho = 1.39$. Any drive then switches both giant fibers on for the rest of the life, producing 618 presses per life. We therefore rescale the gain per circuit so that $\text{gain} \times \rho = 0.8$, which lets activity decay to zero after a stimulus.

Looming inputs. Each flipper is an “eye” at a point on the flipper, and the body receives one of two input variants. In the *single signal* variant, one time-to-contact looming intensity, $\text{gain} \cdot \exp(-\tau/\tau_0)$, drives all input cells of a side, where τ is the time-to-contact in decision steps. In the *split* variant, LC4 cells receive $g_4 \tanh(\dot{\theta}/v_0)$ and LPLC2 cells $g_2 \tanh(\theta/s_0)$, with $\theta = 2 \arctan(R/d)$ and $\dot{\theta} = 2R \text{closing}/(d^2 + R^2)$. This follows the finding that the giant fiber combines an LC4 angular-velocity signal with an LPLC2 angular-size signal [3, 4]. Here d is the distance from the ball to the eye, R an assumed ball radius and “closing” the velocity toward the eye. Both signals are zero unless the ball is closing.

Press logic. On each decision step, each side’s circuit is advanced one step and the giant-fiber activity a is read. A flipper is pressed when a exceeds θ_{on} and it has been released for more than a refractory period of 1–3 steps. It stays pressed until a falls below θ_{off} ($0.7 \theta_{\text{on}}$ for the single-signal body, a ratio sampled in $[0.5, 0.9]$ for the split body), and is force-released after at most 1–3 steps.

Press budget. Presses are free, so a circuit can offset poor selectivity by pressing more. Without a budget, the three tuned shuffles pressed 47.3, 90.3 and 83.3 times per life versus 19.3 for the real wiring. All headline comparisons therefore tune every circuit to at most 25 presses per life on development lives (seeds 7000–7031).

Tuning. The single-signal body searches 60 random configurations per circuit over gain, θ_{on} , τ_0 and hold time, and rejects any with a mean above 25 presses. The split body samples 60 drive configurations plus 50 refinements. In each, θ_{on} is calibrated by a 5-step log-space bisection toward a mean of about 21.5 presses on the first 8 development lives. Survivors are scored on all 32 development lives and accepted only within the budget. Compute is identical for every circuit. Calibrating θ_{on} is essential. Sampling it independently of the gains gave 139.3 presses per life on average over 100 configurations (range 0.0–468.2) and a best dev score of 10.715, against 11.341 with calibration. The press count is a steep, configuration-dependent function of θ_{on} , so independent sampling lands mostly in “never presses” or “spams”.

2.5 Controls and statistics

Shuffled graphs are produced by directed double-edge swaps ($a \rightarrow b$, $c \rightarrow d$ becomes $a \rightarrow d$, $c \rightarrow b$, rejecting self-loops and duplicates), 10 swaps per edge. Node identity, sign, role, channel mapping and every edge weight are retained, and each cell’s in- and out-degree is preserved. About 6.5% of edges remain in place. Each shuffle receives the same training or tuning compute as the real circuit.

Evaluations use 96 paired lives (every policy plays the same seeds). We report mean $\pm$ SE of $\log_{10}(\text{score})$, paired differences with $t = \text{mean difference} / \text{SE}$, and two-sided Wilcoxon signed-rank p -values (normal approximation with tie and zero corrections). With six shuffles the Bonferroni threshold is $0.05/6 = 0.0083$. A pooled test uses the per-life mean over shuffles minus the real circuit. Seeds used to tune or select a checkpoint are never used as its test set. “Presses” are physical flipper down-edges per life, and “c/press” is flipper-ball contacts per press.

3 Results

3.1 Trained brain: no wiring effect

The best brain is the popcode circuit trained by DAgger from the lead1 reflex. Seeds 76000–76095, differences are shuffled minus real:

Wiring	$\log_{10} \pm \text{SE}$	presses	diff vs real	t	Wilcoxon p
real	11.186 ± 0.107	13.1	–	–	–
shuffle 1	11.209 ± 0.094	12.9	$+0.023 \pm 0.092$	+0.25	0.89
shuffle 2	11.240 ± 0.102	13.7	$+0.054 \pm 0.100$	+0.54	0.92
shuffle 3	11.311 ± 0.095	13.8	$+0.125 \pm 0.091$	+1.38	0.2

Table 2: Trained 444-neuron brain on real and shuffled wirings.

Brains trained on shuffled wirings are as good as the one trained on the real wiring. The lead1 teacher itself is only marginally better than the plain reflex in score (11.37 vs 11.28 on fresh seeds 9000–9095, $+0.09 \pm 0.09$, n.s.), although it completes more targets. On seeds 7024–7071 the best brain scores 11.17 ± 0.11 with 13.2 presses. CMA fine-tuning of the popcode seed gave no measurable gain ($+0.05 \pm 0.11$, t 0.5, seeds 6000–6095).

3.2 Giant-fiber body versus six shuffles

Seeds 77000–77095. Every circuit is tuned to at most 25 presses per life, and differences are policy minus real body.

All six shuffle differences have Wilcoxon $p < 0.05$. After Bonferroni, shuffles 1, 5 and 6 remain significant and shuffles 2, 3 and 4 do not. The pooled difference is -0.338 ± 0.079 (t -4.25), or -0.354 ± 0.080 (t -4.43) excluding shuffle 2. The real body makes 0.488 contacts per press against 0.247–0.356 for the shuffles. An earlier run without the press budget (3 shuffles)

Policy	log1p $\pm$ SE	presses	c/press	diff vs real	t	Wilcoxon p
GF body, real	11.221 $\pm$ 0.092	19.3	0.488	–	–	–
shuffle 1	10.795 $\pm$ 0.093	20.5	0.307	-0.426 ± 0.096	-4.45	$7.7\text{e-}5$
shuffle 2	10.968 $\pm$ 0.097	25.4	0.305	-0.253 ± 0.107	-2.37	0.0097
shuffle 3	10.985 $\pm$ 0.098	21.6	0.354	-0.236 ± 0.104	-2.26	0.033
shuffle 4	10.944 $\pm$ 0.100	28.0	0.247	-0.277 ± 0.109	-2.55	0.012
shuffle 5	10.770 $\pm$ 0.091	20.9	0.304	-0.451 ± 0.100	-4.49	$2.6\text{e-}6$
shuffle 6	10.839 $\pm$ 0.102	20.2	0.356	-0.382 ± 0.113	-3.38	0.0026
lead1 reflex	11.480 $\pm$ 0.094	16.4	0.536	$+0.258 \pm 0.105$	$+2.47$	0.0067
never press	9.853 $\pm$ 0.089	0.0	–	-1.368 ± 0.106	-12.95	$7.5\text{e-}16$

Table 3: Giant-fiber body versus six degree-preserving shuffles under the 25-press budget.

shows why the budget matters. The shuffles scored 11.09, 11.39 and 11.23 but pressed 47.3, 90.3 and 83.3 times per life at 0.13 c/press, so they compensated for poor selectivity by pressing 2.5–5x more. The real body stays below the hand-written reflex ($+0.258 \pm 0.105$ for the reflex, p 0.0067) and far above never pressing.

3.3 Split inputs, ablations and shuffles

Seeds 77000–77095. Differences are policy minus split real.

Policy	log1p $\pm$ SE	presses	c/press	diff vs split real	t	Wilcoxon p
split, real	11.242 $\pm$ 0.091	23.7	0.315	–	–	–
LC4 only ($g_2 = 0$)	11.198 $\pm$ 0.107	21.4	0.340	-0.044 ± 0.106	-0.41	0.9
LPLC2 only ($g_4 = 0$)	11.262 $\pm$ 0.097	17.9	0.590	$+0.020 \pm 0.098$	$+0.21$	0.9
single signal, real	11.221 $\pm$ 0.092	19.3	0.488	-0.021 ± 0.093	-0.22	0.78
lead1 reflex	11.480 $\pm$ 0.094	16.4	0.536	$+0.238 \pm 0.093$	$+2.55$	0.023
split shuffle 1	11.098 $\pm$ 0.100	19.0	0.436	-0.143 ± 0.103	-1.39	0.27
split shuffle 2	10.190 $\pm$ 0.085	12.3	0.089	-1.051 ± 0.093	-11.25	$2.2\text{e-}14$
split shuffle 3	10.913 $\pm$ 0.092	28.9	0.230	-0.328 ± 0.095	-3.47	0.00066
split shuffle 4	10.977 $\pm$ 0.089	18.2	0.364	-0.265 ± 0.103	-2.57	0.017
split shuffle 5	10.775 $\pm$ 0.090	16.7	0.320	-0.467 ± 0.101	-4.60	$2.5\text{e-}5$
split shuffle 6	10.820 $\pm$ 0.096	15.2	0.394	-0.421 ± 0.106	-3.99	0.00049

Table 4: Split-input giant-fiber body, ablations and six split shuffles.

The split, LC4-only, LPLC2-only and single-signal bodies are statistically indistinguishable, and all are about 0.24 below the reflex. Four of six split shuffles pass Bonferroni (shuffles 2, 3, 5, 6). Shuffle 1 (p 0.27) and shuffle 4 (p 0.017) do not. The pooled difference is -0.446 ± 0.078 (t -5.73), or -0.325 ± 0.079 (t -4.11) without the outlier shuffle 2, whose circuit achieves only 1.1 contacts per life.

3.4 Spectral radius and latching

The real wiring is markedly more recurrent than its degree-matched shuffles. We probed each circuit with a 10-step step drive on one looming channel (amplitude 0.5), followed by 20 steps of silence:

After the stimulus, all giant-fiber activity returns to 0.000. The real circuit raises the same-side giant fiber 25–40x more than the other side, whereas the shuffles respond on both sides. In the best real configuration the giant fibers are above threshold in 2.8% of steps when the ball is far and 19.5% of steps when it is in the flipper zone ($y > 10$). With the unscaled gain, real and shuffled circuits latched and showed no significant difference.

Circuit	ρ	gain	loom_L drive: GF _L / GF _R	loom_R drive: GF _L / GF _R
real	0.996	0.803	0.316 / 0.008	0.014 / 0.316
shuffle 1	0.776	1.031	0.375 / 0.469	0.373 / 0.450
shuffle 2	0.809	0.989	0.388 / 0.440	0.342 / 0.452
shuffle 3	0.824	0.971	0.456 / 0.405	0.419 / 0.404

Table 5: Response of each circuit to a 10-step drive on one looming channel.

3.5 Gain scan

An 8×8 scan of the (g_4, g_2) gains, with θ_{on} re-calibrated per cell, found 29 of 64 valid cells for the real circuit, 24 for shuffle 1 and 12 for shuffle 2 (cells that cannot be calibrated into the budget are discarded). Low gains were best: the best real cell had g_4 0.50, g_2 0.79 (dev log1p 11.320, 22.4 presses). On 96 evaluation lives that cell scored 11.268 ± 0.099 (22.2 presses). The best real LC4-only cell scored 11.066 ± 0.103 (-0.201 ± 0.105 , $t -1.92$, p 0.14), shuffle 1’s best cell 11.096 ± 0.095 (-0.171 ± 0.109 , p 0.11), and shuffle 2’s best cell 10.323 ± 0.093 (-0.944 ± 0.103 , p 9.2e-13).

3.6 Negative result: analog force

We also tried letting the giant-fiber rate set the flipper force while pressed ($\text{force} = \text{clip}(f_0 + k(a - \theta_{\text{on}}), 0.2, 1.0)$). For the real circuit, the best tuned force was the constant full force ($f_0 = 1$, $k = 0$), so the rate-coded and constant-force bodies are the same policy and score identically (11.221 log1p on the paired evaluation lives). Shuffles with tuned force still lost by 0.29–0.48 (all $p < 0.05$). We dropped the analog trigger, and the flippers remain binary.

3.7 Summary and comparison

Policy	log1p	seeds
GF body (single signal), real	11.221	77000–77095
GF body (split, best cell), real	11.268	77000–77095
lead1 reflex	11.480	77000–77095
trained 444 brain, real	11.186	76000–76095

Table 6: Summary of the headline policies.

Different seed sets are not directly comparable, and no run of the trained brain and the giant-fiber body on the same seeds exists. The statement that they perform similarly rests on 11.19–11.27 versus 11.19 on two different seed sets, with SEs of about 0.1 each.

4 Discussion

Interpretation. The two routes give opposite answers to the same question. When a readout is trained on top of a fixed graph, the trained parameters absorb whatever structure the graph lacks, and real and shuffled wirings are indistinguishable (all $p \geq 0.2$). When nothing is trained and the graph is the body, the wiring has to do the work, and the real wiring does it better than six degree-preserving shuffles by 0.24–0.45 log1p points under a matched press budget. In this task, learning substitutes for wiring, and wiring matters when it cannot be compensated. The advantage lies in press efficiency (0.488 contacts per press vs 0.247–0.356), not in the score ceiling: the real body remains 0.24–0.26 below the hand-written reflex.

Relation to the biology. The giant-fiber result is consistent with the known organisation of the escape pathway. LC4 and LPLC2 each contribute about a tenth to a seventh of the giant fiber’s input and project strictly ipsilaterally, so the real circuit is a lateralised looming detector, and it responds 25–40x more strongly on the same side. The split input scheme follows the proposal that the giant fiber combines LC4 angular-velocity and LPLC2 angular-size signals [3, 4], yet in our task the two channels were interchangeable. We make no claim that pinball play corresponds to escape behaviour in the fly. The looming analogy is an engineering mapping.

Limitations. The framing is post hoc. The giant-fiber pathway was chosen after the trained-readout shuffle control showed no wiring effect, so the hypothesis was formed after that result and after the pathway analysis. The key comparisons were repeated across six shuffles, but all share the same 96 evaluation seeds, which were also used when intermediate variants were compared. The circuit is also a simplified rate network, and ρ is deliberately rescaled to 0.8.

The sensory transduction is hand-designed. Mapping ball kinematics to LC4/LPLC2 drive (time-to-contact, or angular size and velocity from a single point per flipper with a hand-chosen R) is engineering, not a measured receptive field, and the v1.0 annotations contain no usable retinotopic coordinates. A few constants are searched separately per circuit (4 for the single-signal body, and R , g_4 , g_2 , v_0 , s_0 and θ_{on} for the split body). The press budget and per-circuit calibration are part of the protocol, but differences in contacts per press may partly reflect how well each circuit can be calibrated, since some shuffles had only 12–24 valid scan cells out of 64.

There is no side-preserving shuffle. Only degree-preserving shuffles were run, and these destroy ipsilateral structure, so the advantage may come partly from side labelling of inputs rather than finer wiring. A control that rewires while keeping which side each edge sits on has not been done. Only six shuffles of a single circuit and one input-to-GF design were tested.

A one-threshold reflex scores 11.3–11.5 with about 12–16 presses per life, and every non-aiming policy sits at about 11.3–11.45. Both brains are at or below this ceiling, and neither aims. The DAgger teacher is a hand-designed reflex, and the trained brain ties it. The trained brain and the giant-fiber body have also never been compared on the same seeds.

The body has a blind spot. A diagnosis of the body (default parameters) over 9,484 recorded down-moving states shows that 4.1% are blind (both normalised LC4 and LPLC2 drives below 0.25), 7.3% in the flipper zone and 16.8% in the centre gap ($|x| < 0.96$). Over 64 lives, 60 ended in a drain and 53 (88.3%) were “unseen”: the ball was in flipper reach in the last 20 steps with no press and the giant fibers below threshold. The cause is a single receptive point per flipper, so a ball falling through the centre gap or past a tip has almost no closing component toward either point. A retinotopic fix, giving every LC4/LPLC2 cell its own receptive-field point along the flipper sweep, is under development and not evaluated here.

Statistical power is modest. SEs are about 0.1 log1p on 96 lives, so effects of 0.2 are borderline, and the pooled shuffle test is the stronger summary. No full-game numbers exist for the giant-fiber body.

Future work. The immediate steps are a side-preserving shuffle control, the retinotopic body to close the blind spot, a held-out evaluation of an online learner of the five drive parameters (a partial run was lost before final evaluation), and a same-seed comparison of the trained and untrained brains. Beyond that, aiming, which neither brain does, is the main route above the reflex ceiling.

5 Conclusion

A fixed circuit built from the *Drosophila* looming-escape pathway, with no training, plays pinball better than six degree-preserving shuffles of itself under a matched press budget (pooled -0.338 ± 0.079 , $t = -4.25$). A trained readout on a generic 444-neuron subnetwork shows no wiring effect at all. The wiring helps in efficiency, not in ceiling: both brains remain at or below a hand-written reflex. Given the post hoc framing, the hand-designed sensory transduction and the

missing side-preserving control, the results are suggestive rather than conclusive. They do point to a simple rule: wiring matters when nothing is trained.

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Data and code availability

Code is available in the project repository (DrosophilaCadet) under the MIT license. Connectome data are MaleCNS v1.0 (CC-BY 4.0), downloadable without login. The game data, sounds and splash of 3D Pinball for Windows – Space Cadet are copyright Cinematronics / Maxis, now Electronic Arts, and are not distributed with the repository except as noted in its copyright section; the engine is the MIT-licensed decompilation by k4zmu2a [2]. This project is an unofficial fan project.